

Achilles Tendon Rupture Rehabilitation

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The medical text below was not translated, paraphrased, or supplemented.

Important:

Please read before commencing any exercise:

You are most vulnerable to re injury in the first 8 weeks after coming out of the boot as this is when the tendon is at its weakest. If you do not follow advice given by your therapist or try to carry out exercises/ activities that have not been recommended, you are at risk of re injury. This may lead to poor longer-term outcomes for your recovery.

You must not bring your foot towards you (pull your toes back) unless directed by your physiotherapist

What is the Achilles tendon?

The Achilles tendon is the strongest tendon in your body, it connects your calf muscles (Gastrocnemius and soleus) to your heel bone.

When your calf muscle contracts it allows you to point your toes and produce enough force to walk, run and jump. The Achilles helps to make these movements even more efficient by storing energy when it stretches and releases this energy. It acts a little bit like a spring.

When you bring your foot towards you this stretches the Achilles and calf which is something you'll be avoiding in the early stages of rehab.

Early Stages: Phase 1 to 2 (Injury to 8 weeks)

Research has shown that there is very little to no difference in outcomes and re-rupture rates between surgical and non-surgical management of Achilles rupture.

The accepted default management of an Achilles tendon rupture is conservative treatment. This means a non-surgical approach.

This begins with immobilisation of the foot in a position with your foot pointing downwards. This ensures that the ends of the tendon have a good opportunity to join and start to heal together and are not stretched apart. Stretching the tendon will impede this process.

Over time the foot is brought to a plantigrade (flat) position.

At Derriford Hospital you will spend 8-9 weeks in the Vacoped boot which is issued to you. This will support your lower leg and ankle while it heals. It will also let you start to put weight through the leg sooner and we would encourage that.

We would advise that you are in the boot no longer than 8 weeks in total.

You will need to be on a blood thinner injection daily to prevent clots which you should be issued with for 6 weeks.

At week 6 you will be allowed to go to bed without the boot on. You can stop the injections, and you should bring the used needles and any outstanding ampoules back to the hospital for disposal.

Time frame

Boot information

ED/Fracture Clinic

Into Vacoped boot fixed at Level 3 for 3 weeks from fitting of the boot. Full weightbearing as pain allows with elbow crutches. Heel raises issued.

Week 3 from date of boot fitting

Adjust the boot (bottom screw) by 1 notch to 2.5 and then 1 notch every 5 days until the boot is at 0 (Flat Piece on now) Don not adjust the top screw .

Adjust the boot (bottom screw) by 1 notch to 2.5 and then 1 notch every 5 days until the boot is at 0 (Flat Piece on now)

Don not adjust the top screw .

Level 0 (Around Week 6)

Boot adjusted to 0 with flat piece on bottom. Boot can be removed at night whilst sleeping. Must be put back on when weightbearing. Stop injections.

Week 6 to 7

Further 2 weeks of wearing boot at Level 0 with flat piece on (No further adjustments).

Week 8 to 9 Onwards

Boot can now be removed if under the care of physiotherapy or referral made and can now fully weight-bear in normal shoe with heel raises.

Adjustments and timeframe

Boot setting

Date of boot fitting

(Level 3) Heel Raises to be issued

(Level 3)

Heel Raises to be issued

1st Adjustment (3 weeks)

(Level 2.5)

2nd Adjustment (5 days later)

(Level 2)

3rd Adjustment (5 days later)

(Level 1.5)

4th Adjustment (5 days later)

(Level 1)

5th Adjustment (5 days later)

(Level 0.5)

6th Adjustment (5 days later)

(Level 0 and flat piece on bottom of boot)

You will be reviewed twice in the Foot and Ankle Clinic during your recovery.

Recovery timeline

Healing is measured from the date of boot fitting

Full recovery can take a long time and should be guided by your Physiotherapist and not rushed. It may take up to 18 to 24 months sometimes longer.

Phase and timeframe

Description of phase

Phase 1 Injury to 2 weeks

Phase 1

Injury to 2 weeks

Plaster of Paris Cast with foot pointing downwards.

Phase 2 3 weeks to 8 weeks

Phase 2

3 weeks to 8 weeks

Vacoped boot gradually adjusted over 8 weeks with guidance. Boot removed over night at 6 weeks.

Phase 3 8 weeks to 16 weeks

Phase 3

8 weeks to 16 weeks

Gentle non-weightbearing exercises, progressing to weightbearing exercises all without pulling toes towards you.

Phase 4 4 months to 6 months

Phase 4

4 months to 6 months

Gentle jogging on the spot, normal walking, single leg exercises.

Phase 5 6-12 months

Phase 5

6-12 months

Balance exercises, sport/activity specific training

Phase 6 8-12 months

Phase 6

8-12 months

Running, jumping and full return to sport.

Expectations:

It is likely to be 12 weeks before you can drive (Unless it is your (Left) leg and you have an automatic car). You need to be able to safely perform an emergency stop. It is likely to be 6 months before you can run, climb ladders or do lower limb gym work.

Vacoped boot Information

While you are encouraged to remove the boot to check skin condition, do not be tempted to weight-bear through the foot without the boot on. You are very safe whilst in the boot and can fully weight-bear as able.

You must not bring your foot towards you (do not pull your toes back towards your shin)

Boot Instructions

Wear the boot in bed until you reach Level 0 (You may take off the bottom wedge when it is on in bed but put it back on when getting up.)

The boot can be removed for a shower if you are able to stand safely without putting weight through the affected leg. Plan this before you have a shower.

Remove the boot daily to check the skin condition.

Move the beads within the boot regularly (daily) to prevent chafing of the skin.

Change from the wedge sole to the flat sole at Level 0.

Do not adjust the top screw of the boot at any point

Watch this video for boot fitting instructions

View the boot fitting pdf poster

Help and Advice

It is not uncommon for patients to have issues with chafing or rubbing on the skin around the ankle. If it is a problem, allowing some air into the boot liner to reduce the stiffness of the liner is permitted. It may be useful to shake the liner out to redistribute the beads regularly.

If you have concerns about the boot, your skin or any issues pertaining to the process please contact us on the phone number below.

It is likely that you will have occasional twinges and aches as your Achilles continues to heal. Any pulling sensation in the back of the heel/calf may indicate that your tendon is healing and could be an encouraging rather than worrying sign.

You will have been given medication to prevent Deep Vein Thrombosis (DVT) and advised on what signs and symptoms to look out for by the fracture clinic staff please take it as instructed.

Deep Vein Thrombosis (DVT) Signs and Symptoms

Sudden cramping in the affected calf.

Significant redness or heat on the affected leg.

Sudden increase in pins and needles.

New chest pain or shortness of breath.

If you experience any symptoms related to Deep Vein Thrombosis (DVT) then seek urgent medical attention.

Contacts

Fracture Clinic: 01752 430905 (Monday to Friday, 8:30am to 4:30pm). Go to ED or UTC Cumberland Centre outside of these times.

General queries: Email: plh-tr.orthopaedicfootandankle@nhs.net

Exercises

Exercises can be started in the Vacoped boot (Quadriceps and Hamstring Strengthening)

Exercises at Weeks 3-6

View the PDF for exercise images

Static quad exercises

Either sitting or lying down, tighten the muscles on the front of your thigh by pushing the back of your knee down into the bed (see Figure 1).

Keep your leg straight on a flat surface.

Hold for five to 10 seconds, then relax.

Inner range quad exercises

Lie down and place a rolled towel under your knee for support.

Tighten your thigh muscle and lift your foot up to straighten your knee. Keep your knee on the towel at all times.

Hold for 5 to 10 seconds, then relax.

Straight leg raises

With your leg straight on a flat surface, tighten your thigh muscles to hold your knee straight and lift your leg up. Aim to lift your leg 30 to 40cm off the surface then lower it down gently.

Keep your knee straight at all times.

Seated knee extension exercises

Sitting down with your legs hanging over the edge of a bed or chair (see figure 6), raise your foot until you straighten your leg (see figure 7).

Hold for 5 to 10 seconds then lower it down gently.

Prone knee bend exercise

Lay flat on your front (see figure 8).

Slowly bend your knee up to 90° and lower it back down again (see figure 9). If you find this too easy, speak to your physiotherapist as they can make this exercise harder by adding a resistance band.

Hip extension in prone

Lay on your front (see figure 10).

Tighten your bottom muscles and lift your thigh up off the floor (see figure 11). Keep your knee straight at all times. Do not allow your pelvis to lift off the floor.

Hold for 5 to 10 seconds then lower down gently.

Standing exercises in Vacoped boot

Stand up and hold on to something sturdy for support (for example your kitchen counter). Make sure you keep your back straight with every exercise.

Standing knee flexion

Stand with both legs straight and together.

Bend your knee, bringing your foot up behind you.

Keep your hips straight and your knees together.

Relax back down again and repeat.

Hip abduction in standing

Standing up tall with your feet together, bring your leg out to the side whilst keeping your knee straight.

Keep your body straight, do not lean to the side as you lift your leg.

Gently lower your leg back down again.

If too easy, progress to side lying (see next exercise).

Hip abduction inside lying

Laying on your side, lift your top leg up towards the ceiling.

Keep your knee straight at all times.

Your hip should be in line with your knee and ankle. Do not allow your hips to roll back as you lift your leg.

Your aim at week 6 is to have a full range of movement in your hip and knee, and no pain in your Achilles tendon while you are resting.

Phase 2 (exercises at 4 to 8 weeks)

At week 6, you will be allowed to remove the Vacoped boot overnight and when doing the following exercises.

View the PDF for exercise images

Calf contractions with holds

Put a small ball or rolled towel against a wall and gently press the ball of your foot into the ball for 10-20 seconds. Do not flex the knee or stretch the calf.

Seated heel raises

Sit straight on a chair with your feet flat on the floor. Raise your heels off the ground and push through your toes. Hold 5-10 seconds and then slowly lower.

You must not bring your foot towards you (pull your toes back) unless directed by your physiotherapist

Phase 3 (exercises at 8 to 16 weeks)

At week 8 you should commence Physiotherapy; we would expect you to be getting into a normal shoe with a heel raise in it, within a couple of weeks, which you would wear until about 6 months after the injury.

We will give you some heel raises at the fracture clinic appointment.

During this phase your Physiotherapist will guide you through exercises and gradually look at range of movement and strengthening of the ankle. You may also commence balance exercises to regain the necessary stability in the ankle.

Do not force dorsiflexion or calf stretches.

Be cautious when coming downstairs.

Your exercises will be progressed by your physiotherapist.

No Running, no Jumping, no Hopping, no Hill or brisk walking.

Speak to your physiotherapist before returning to the gym.

Phase 4 (exercises at 4 to 6 months)

During this phase you work towards removal of the heel raise in your shoe.

You must continue to refrain from activity that put your calf under strain while in a dorsiflexed position e.g., explosive change of direction or heel drops of a step.

Your Physiotherapist will direct you based on the stage you are at and how you are progressing.

You may well begin single leg heel raises from the floor and small amounts of 2-foot jumping.

No Running

Phase 5 (exercises at 6 to 12 months)

In Phase 5 you will be guided by your Physiotherapist to work towards more specific strengthening and sports specific activity.

You may start jogging on the flat and begin to work on eccentric strengthening with heel raises of the step.

Towards the end of this Phase, it is likely that a return to sport plan is developed and your Physiotherapist will be able to use certain tests to establish your readiness for a return to training and competition if that is your goal.

Your progress

In all the phases of your rehabilitation please follow the instruction and advice given by the members of the team, whether that be in fracture clinic or your Physiotherapist.

Do not compare your progress to that of anyone else who has seemingly been through a similar injury. Everyone's journey is different, and it is very important that you follow the guidance of the trained health care professionals looking after you.

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